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```
clc;
clear;
close all;
```

PARAMETERS

```
M = 16;
k = log2(M);
```

```
Nfft = 64;
Nofdm = 100;
```

```
SNR_dB = 0:5:25;
```

```
BER = zeros(size(SNR_dB));
```

TRANSMITTER

```
Nbits = Nofdm * Nfft * k;
```

```
txBits = randi([0 1], Nbits, 1);
```

```
% 16-QAM modulation
txSymbols = qammod(txBits, M, ...
    'InputType', 'bit', ...
    'UnitAveragePower', true);
```

```
% Arrange symbols into OFDM symbols
txSymbolsMatrix = reshape(txSymbols, Nfft, Nofdm);
```

```
% IFFT
txOFDM = ifft(txSymbolsMatrix, Nfft);
```

AWGN CHANNEL + RECEIVER

```
for i = 1:length(SNR_dB)
```

```
    % Add AWGN
    rxOFDM = awgn(txOFDM, SNR_dB(i), 'measured');
```

```
    % FFT
```

```

rxSymbolsMatrix = fft(rxOFDM, Nfft);

% Convert to serial
rxSymbols = rxSymbolsMatrix(:);

% 16-QAM demodulation
rxBits = qamdemod(rxSymbols, M, ...
    'OutputType', 'bit', ...
    'UnitAveragePower', true);

% BER
[numErrors, BER(i)] = biterr(txBits, rxBits);

fprintf('SNR = %2d dB | Errors = %d | BER = %e\n', ...
    SNR_dB(i), numErrors, BER(i));

end

SNR = 0 dB | Errors = 7326 | BER = 2.861719e-01
SNR = 5 dB | Errors = 4118 | BER = 1.608594e-01
SNR = 10 dB | Errors = 1530 | BER = 5.976563e-02
SNR = 15 dB | Errors = 135 | BER = 5.273438e-03
SNR = 20 dB | Errors = 0 | BER = 0.000000e+00
SNR = 25 dB | Errors = 0 | BER = 0.000000e+00

```

BER GRAPH

```

figure;

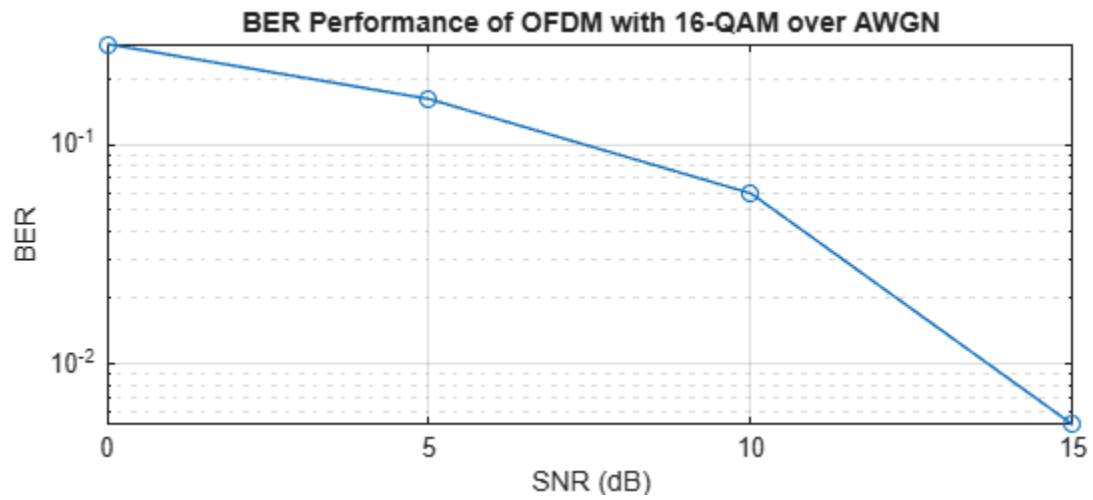
semilogy(SNR_dB, BER, '-o');

grid on;

xlabel('SNR (dB)');
ylabel('BER');

title('BER Performance of OFDM with 16-QAM over AWGN');

```



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